IGCSE 0607 Practice Questions — E2.5 — Combined Questions & Markscheme

Equations

Total Questions: 6 — **Total Marks:** 44 Marks

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1 **(a)** Solve the equation.

$$4(2x - 1) = 3x + 11$$

$x =$ ..... **[3]**

(b) Make x the subject of the formula.

$$y = \frac{3x - 5}{2}$$

$x =$ ..... **[3]**

2 Consider the following algebraic tasks.

(a) Solve the equation.

$$5(x-2) = 3x + 14$$

$x =$ ..... **[3]**

(b) Rearrange the formula to make b the subject.

$$A = \frac{h(a+b)}{2}$$

$b =$ ..... **[3]**

3 Solve the following equations.

(a) Solve the simultaneous equations.
You must show all your working.

$4x + 3y = 5$

$2x - 5y = 9$

$x =$ ..... , $y =$ ..... **[4]**

(b) Solve the equation by factorisation.

$3x^2 - 10x - 8 = 0$

$x =$ ..... or $x =$ **[4]**

4 **(a)** A rectangle has length $(3x + 2)$ cm and width $(x - 1)$ cm.
The perimeter of the rectangle is 42 cm.

Work out the value of x.

$x =$ ..... **[3]**

(b) Make x the subject of the formula.

$$p = \sqrt{\frac{x-3}{x}}$$

$x =$ ..... **[4]**

5 **(a)** Show that $\frac{3}{x-1} - \frac{2}{x+3} = 1$ simplifies to $x^2 + x - 14 = 0$.

[4]

(b) Solve the equation $x^2 + x - 14 = 0$.
Show all your working and give your answers correct to 2 decimal places.

$x =$ ..... , $x =$ ..... **[3]**

6 Consider the following algebraic equations and formulas.

(a) Rearrange the formula to make x the subject.

$$y = \frac{3x - 1}{x + 2}$$

$x =$ ..... **[4]**

(b) Solve the equation.
Give your answers correct to 2 decimal places.
Show all your working.

$$\frac{2}{x - 1} + \frac{3}{x + 2} = 1$$

$x =$ ..... or $x =$ **[6]**

1 **(a)** $8x - 4 = 3x + 11$ **[M1]**

$5x = 15$ **[M1]**

3 **[A1]**

(b) $2y = 3x - 5$ **[M1]**

$2y + 5 = 3x$ **[M1]**

$\frac{2y+5}{3}$ or $\frac{2y}{3} + \frac{5}{3}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

2 **(a)** $5x - 10 = 3x + 14$ **[M1]**

$2x = 24$ **[M1]**

12 **[A1]**

(b) $2A = h(a + b)$ or $\frac{A}{h} = \frac{a+b}{2}$ **[M1]**

$\frac{2A}{h} = a + b$ or $2A = ha + hb$ **[M1]**

$\frac{2A}{h} - a$ or $\frac{2A - ah}{h}$ **[A1]**

[Back to Question] *[Total Marks: 6]*

3 **(a)** Multiply second equation by 2: $4x - 10y = 18$ **[M1]**

Subtract equations to eliminate x: $13y = -13$ **[M1]**

$y = -1$ **[A1]**

$x = 2$ **[A1]**

(b) $(3x + 2)(x - 4) = 0$ **[M2]**

If M0, M1 for $(3x \pm a)(x \pm b)$ where $ab = 8$ or $a \pm 3b = -10$

$x = -\frac{2}{3}$ **[A1]**

$x = 4$ **[A1]**

[Back to Question] *[Total Marks: 8]*

4 **(a)** $2(3x + 2 + x - 1) = 42$ or $3x + 2 + x - 1 = 21$ **[M1]**

$4x + 1 = 21$ **[M1]**

5 **[A1]**

(b) $p^2 = \frac{x-3}{x}$ **[M1]**

$p^2 x = x - 3$ **[M1]**

$p^2 x - x = -3$ or $x - p^2 x = 3$ or $x(p^2 - 1) = -3$ **[M1]**

$\frac{3}{1-p^2}$ or $\frac{-3}{p^2-1}$ **[A1]**

[Back to Question] *[Total Marks: 7]*

5 (a) $\frac{3(x+3)-2(x-1)}{(x-1)(x+3)} = 1$ **[M1]**

$3x+9-2x+2 = (x-1)(x+3)$ **[M1]**

$x+11 = x^2+2x-3$ **[M1]**

$x^2+x-14 = 0$ established with no errors seen **[A1]**

(b) $\frac{-1 \pm \sqrt{1^2-4(1)(-14)}}{2(1)}$ **[M1]**

$\frac{-1 \pm \sqrt{57}}{2}$ **[M1]**

3.27, −4.27 **[A1]**

[Back to Question] *[Total Marks: 7]*

6 (a) $y(x+2) = 3x-1$ **[M1]**

$yx+2y = 3x-1$ **[M1]**

$2y+1 = 3x-yx$ or $yx-3x = -2y-1$ **[M1]**

$x = \frac{2y+1}{3-y}$ or $x = \frac{-2y-1}{y-3}$ **[A1]**

(b) Multiply throughout by $(x-1)(x+2)$: $2(x+2)+3(x-1) = (x-1)(x+2)$ **[M1]**

Expand terms: $2x+4+3x-3 = x^2+x-2$ **[M1]**

Collect to form a quadratic equation: $x^2-4x-3 = 0$ **[A1]**

Use the quadratic formula: $x = \frac{-(-4) \pm \sqrt{(-4)^2-4(1)(-3)}}{2(1)}$ **[M1]**

$x = \frac{4 \pm \sqrt{28}}{2}$ **[M1]**

$x = 4.65$ and $x = -0.65$ **[A1]**

[Back to Question] *[Total Marks: 10]*